

1. Input Semantics Prompt Patterns

This category deals with how a Large Language Model (LLM) understands the input and translates it into something it can use to generate output.

Prompt Name	When to Use It (Intent & Motivation)	Example
Meta Language Creation	Intent: To create the prompt via an alternate language, such as a textual shorthand notation for graphs, state transitions, or commands. Motivation: Conventional human language may be ill-suited for concisely, unambiguously, or clearly expressing complex problems or structures. This pattern teaches the LLM the necessary semantics for the new language.	“From now on, whenever I type two identifiers separated by a ' $\rightarrow$ ', I am describing a graph. For example, ' $a \rightarrow b$ ' is describing a graph with nodes 'a' and 'b' and an edge between them. If I separate identifiers by ' $-[w:2, z:3]\rightarrow$ ', I am adding properties of the edge, such as a weight or label.”

2. Output Customization Prompt Patterns

This category focuses on constraining or tailoring the types, formats, structure, or other properties of the output generated by the LLM.

Prompt Name	When to Use It (Intent & Motivation)	Example

Output Automater	Intent: To have the LLM generate a script or other automation artifact that can automatically perform any steps it recommends, reducing the manual effort required from the user. Motivation: Having users continually perform manual, error-prone steps dictated by LLM output is tedious. This pattern translates steps into an executable artifact.	“From now on, whenever you generate code that spans more than one file, generate a Python script that can be run to automatically create the specified files or make changes to existing files to insert the generated code.”
Persona	Intent: To give the LLM a specific point of view, role, perspective, or character ("persona") that helps it select what details and types of output to generate. Motivation: Users often know the <i>role</i> or <i>type of person</i> they would ask for help but may not know the exact output or details required from that expert.	“From now on, act as a security reviewer. Pay close attention to the security details of any code that we look at. Provide outputs that a security reviewer would regard regarding the code.”

Visualization Generator	Intent: To use text generation to create inputs for external visualization tools (like Graphviz Dot or DALL-E) to produce diagrams or images, as many concepts are easier to grasp visually. Motivation: LLMs generally produce text and cannot draw diagrams directly. This pattern overcomes that limitation by generating input text in the correct format for visualization tools.	“Whenever I ask you to visualize something, please create either a Graphviz Dot file or DALL-E prompt that I can use to create the visualization. Choose the appropriate tools based on what needs to be visualized.”
Recipe	Intent: To output a complete sequence of steps to achieve a stated goal, given some partially provided information or "ingredients" that define constraints or intermediate steps. Motivation: Users often know the end goal and some components (ingredients) but need the LLM to fill in the knowledge gaps, such as the precise ordering or missing steps.	“I am trying to deploy an application to the cloud. I know that I need to install the necessary dependencies on a virtual machine for my application. I know that I need to sign up for an AWS account. Please provide a complete sequence of steps. Please fill in any missing steps. Please identify any unnecessary steps.”

Template	Intent: To ensure the LLM's output follows a precise, specified structure or format using placeholders that the LLM is instructed to fill with content. Motivation: Output sometimes must be produced in an exact format that is application- or use-case specific and not known inherently by the LLM, requiring explicit instruction on structure.	"I am going to provide a template for your output. Everything in all caps is a placeholder. Any time that you generate text, try to fit it into one of the placeholders that I list. Please preserve the formatting and overall template that I provide at https://myapi.com/NAME/profile/JOB"
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3. Error Identification Prompt Patterns

This category focuses on identifying and helping to resolve errors or inaccuracies in the output generated by the LLM.

Prompt Name	When to Use It (Intent & Motivation)	Example
Fact Check List	Intent: To ensure the LLM outputs a list of facts or assumptions that the answer depends on, so the user can perform due diligence and validate the veracity of the output. Motivation: LLMs often generate factually incorrect text (hallucinations). This pattern flags important facts for verification, reducing the reliance on the tool's convincing but potentially flawed outputs.	"From now on, when you generate an answer, create a set of facts that the answer depends on that should be fact-checked and list this set of facts at the end of your output. Only include facts related to cybersecurity."

Reflection	Intent: To ask the model to automatically explain the reasoning and assumptions behind its answers. This helps users assess the output's validity, debug their own prompts, and understand how the LLM arrived at a particular answer. Motivation: LLMs can provide incomplete, incorrect, or ambiguous answers. Reflection helps address these shortcomings and is particularly effective when dealing with nuanced topics where interpretation matters.	"When you provide an answer, please explain the reasoning and assumptions behind your selection of software frameworks. If possible, use specific examples or evidence with associated code samples to support your answer of why the framework is the best selection for the task. Moreover, please address any potential ambiguities or limitations in your answer, in order to provide a more complete and accurate response."
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4. Prompt Improvement Prompt Patterns

This category focuses on improving the quality of the user's input and the resultant output.

Prompt Name	When to Use It (Intent & Motivation)	Example
Question Refinement	Intent: To engage the LLM in the prompt engineering process by automatically suggesting potentially better or more refined versions of the user's question, aiding the user in finding the right query. Motivation: The user may not be an expert or know the best way to phrase the question. The LLM can use its knowledge to suggest a revised prompt that incorporates necessary context or specifics.	"From now on, whenever I ask a question about a software artifact's security, suggest a better version of the question to use that incorporates information specific to security risks in the language or framework that I am using instead and ask me if I would like to use your question instead."

Alternative Approaches	Intent: To ensure the LLM always offers alternative ways of accomplishing a user-specified task, preventing the user from pursuing only familiar or potentially suboptimal approaches. Motivation: Humans often suffer from cognitive biases or may be unaware of better or novel solutions. This pattern forces the presentation of alternatives for informed decision-making.	“Whenever I ask you to deploy an application to a specific cloud service, if there are alternative services to accomplish the same thing with the same cloud service provider, list the best alternative services and then compare/contrast the pros and cons of each approach with respect to cost, availability, and maintenance effort and include the original way that I asked. Then ask me which approach I would like to proceed with.”
Cognitive Verifier	Intent: To force the LLM to always subdivide complex questions into a series of smaller, component questions, the answers to which are then combined to produce a more accurate overall answer. Motivation: Addresses questions that are initially too high-level and leverages research showing that LLMs can reason better and produce more accurate responses when the problem is explicitly broken down into smaller steps.	“When I ask you a question, generate three additional questions that would help you give a more accurate answer. When I have answered the three questions, combine the answers to produce the final answers to my original question.”

Refusal Breaker	Intent: To ask an LLM to automatically help users rephrase a question when it refuses to provide an answer (e.g., due to policy or lack of information). Motivation: Refusals can frustrate users. This pattern provides both an explanation for the refusal and alternative, acceptable wordings, giving the user a path forward.	“Whenever you can’t answer a question, explain why and provide one or more alternate wordings of the question that you can’t answer so that I can improve my questions.”
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5. Interaction Prompt Patterns

This category focuses on redefining the interaction flow between the user and the LLM.

Prompt Name	When to Use It (Intent & Motivation)	Example
Flipped Interaction	Intent: To have the LLM ask questions to obtain the information needed to perform a task, thereby flipping conversational control from the user to the LLM. Motivation: The LLM can often use its inherent knowledge to select the format and content of interactions more accurately and efficiently to achieve the specified goal, such as gathering deployment parameters.	“From now on, I would like you to ask me questions to deploy a Python application to AWS. When you have enough information to deploy the application, create a Python script to automate the deployment.”

Game Play	Intent: To create a game centered around a specific topic, with the LLM generating output in the form of a game, often acting as the game master. Motivation: Enables the generation of scenarios or questions revolving around a topic that requires problem-solving (like cybersecurity investigation), automating the creation of wide-ranging game content that would be time-consuming to generate manually.	“We are going to play a cybersecurity game. You are going to pretend to be a Linux terminal for a computer that has been compromised by an attacker... I am going to use commands to try and figure out how the system was compromised... To start the game, print a scenario of what happened that led to my investigation and make the description have clues that I can use to get started.”
Infinite Generation	Intent: To automatically generate a series of outputs indefinitely or repetitively without the user needing to re-enter the generator prompt each time, minimizing user input. Motivation: Many tasks (such as generating multiple instances of data or code based on a formula) require the repetitive application of the same prompt, which is tedious and error-prone if the user must retype the prompt constantly.	“From now on, I want you to generate a name and job until I say stop. I am going to provide a template for your output. Everything in all caps is a placeholder... Please preserve the formatting and overall template that I provide: https://myapi.com/NAME/profile/JOB”

6. Context Control Prompt Patterns

This category focuses on controlling the contextual information within which the LLM operates.

Prompt Name	When to Use It (Intent & Motivation)	Example
Context Manager	Intent: To enable users to explicitly specify (include) or remove (exclude) context for a conversation, focusing the LLM on specific topics or excluding irrelevant prior dialogue. Motivation: LLMs often struggle to interpret the intended context or focus due to prior inputs or irrelevant attention on the wrong statements, so explicit management ensures relevance and coherence.	To Specify Context: “When analyzing the following pieces of code, only consider security aspects.” To Remove Context: “When analyzing the following pieces of code, do not consider formatting or naming conventions.”